CCH® Tagetik | **ACADEMY**

Analytical Workspace designer



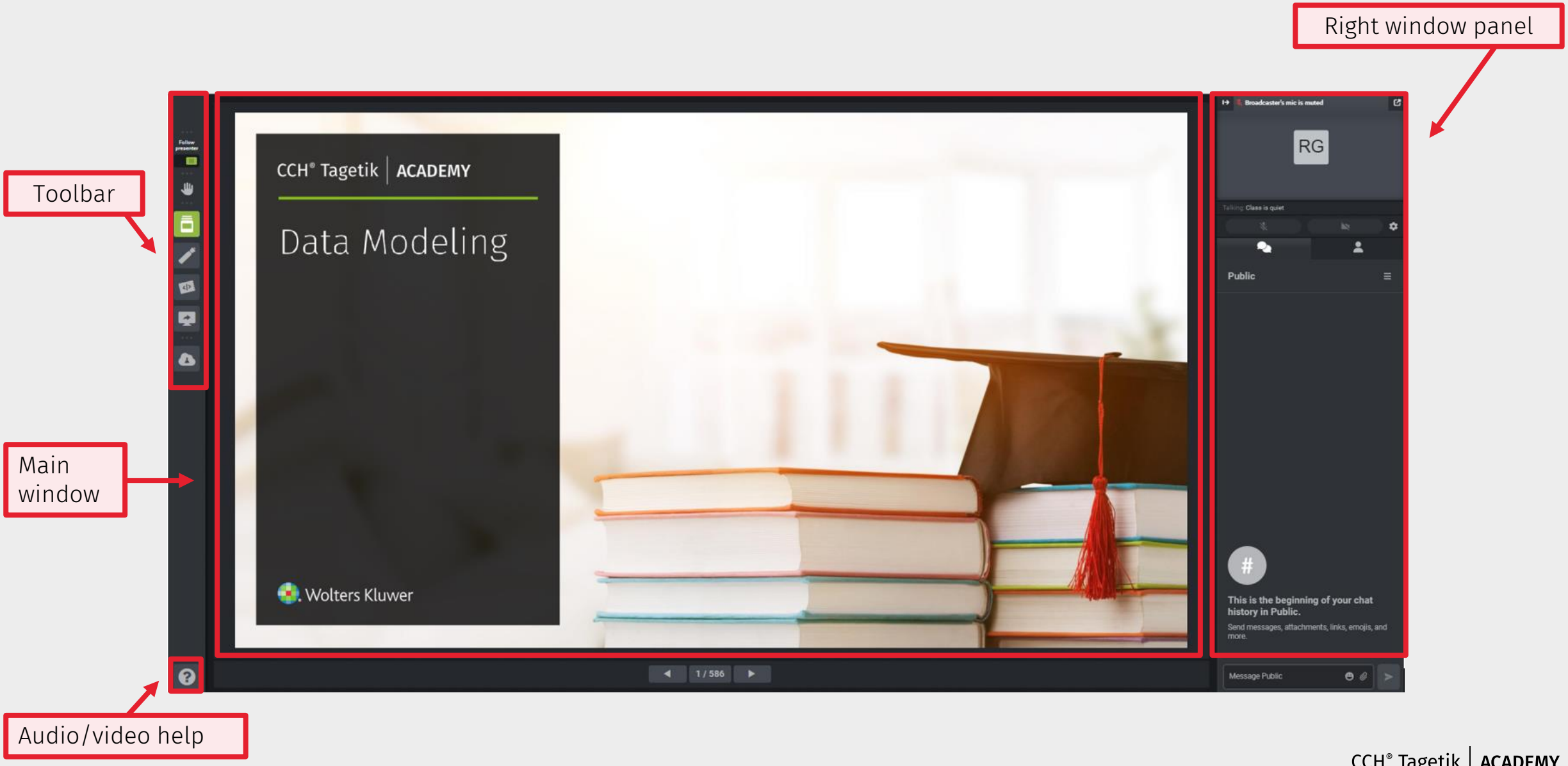
Wolters Kluwer

Before starting

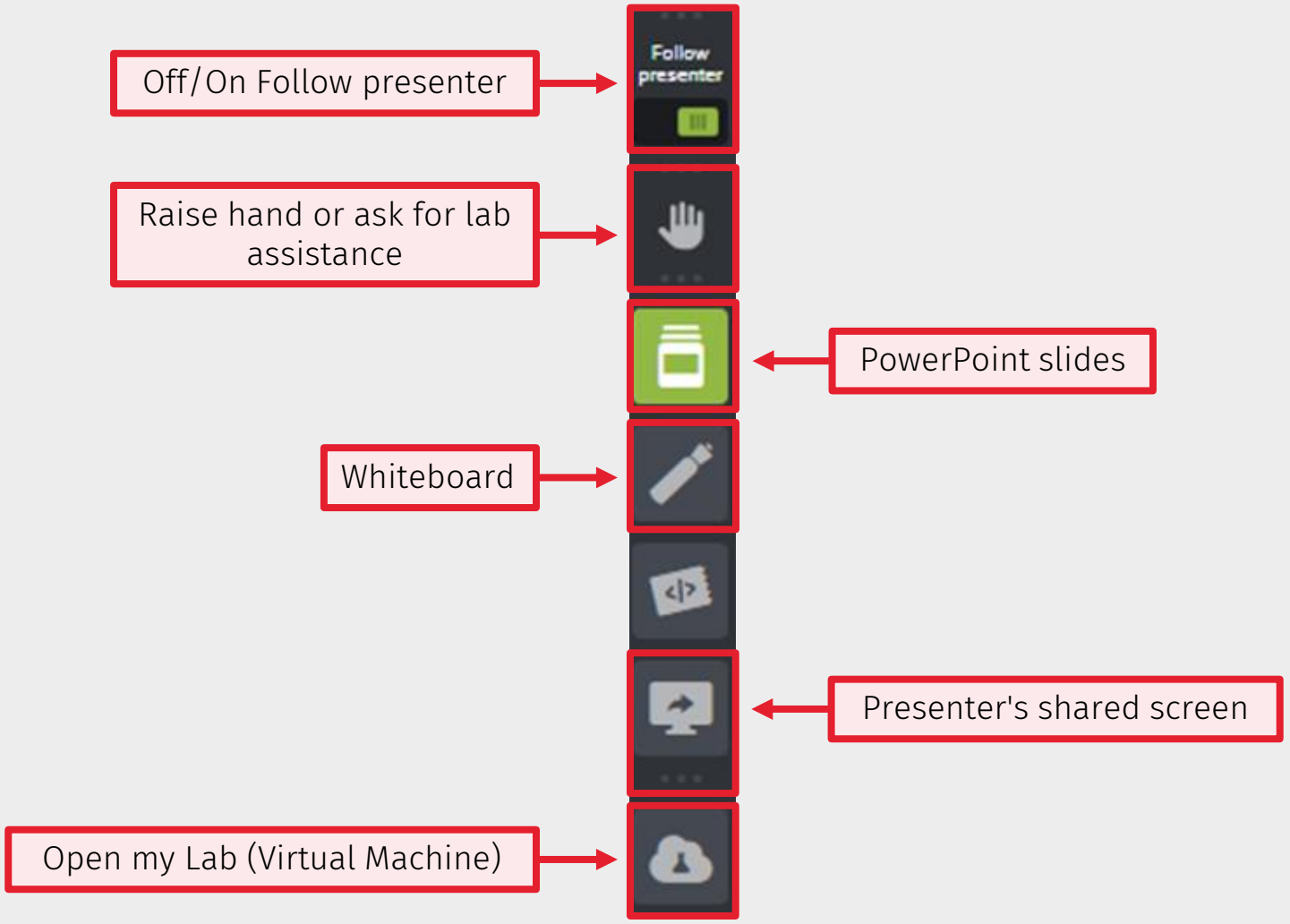
Strigo platform - “How to”



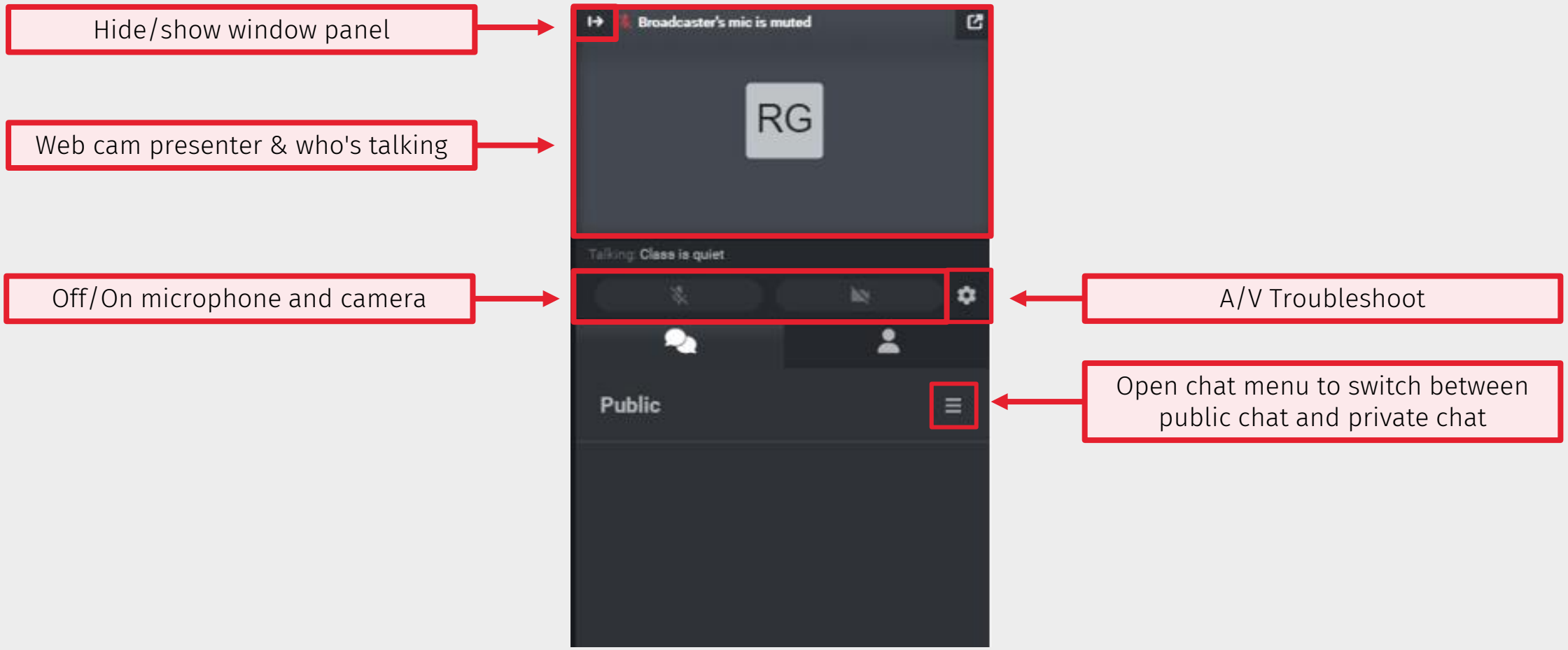
Strigo overview (for Instructor-led classes)



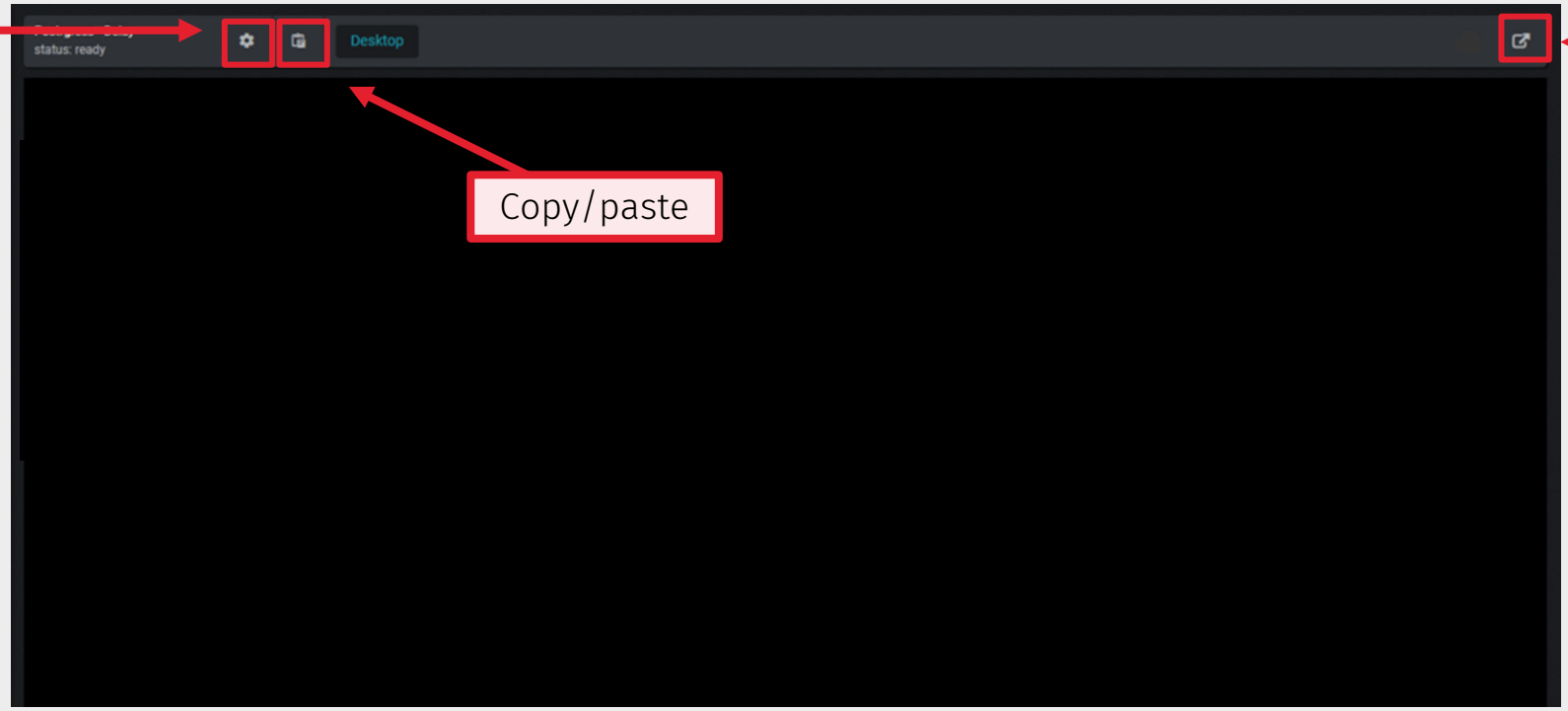
Strigo left toolbar



Strigo right window panel



Strigo Lab (Virtual Machine)

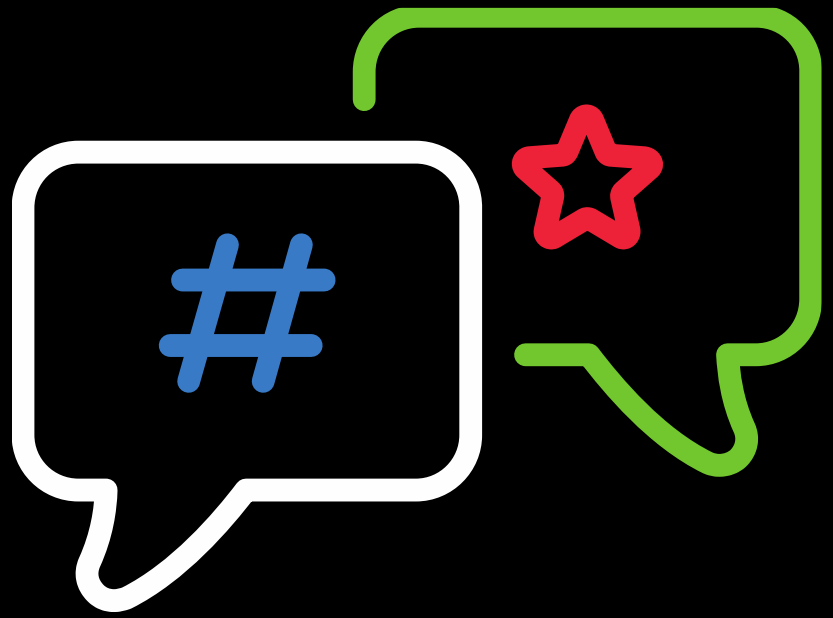


Agenda

- Data model and design consideration
- Discussion on solutions to specific needs
- Solution design workshops
- Discussion on solution design
- Hands-on exercises
- Technical design guidelines



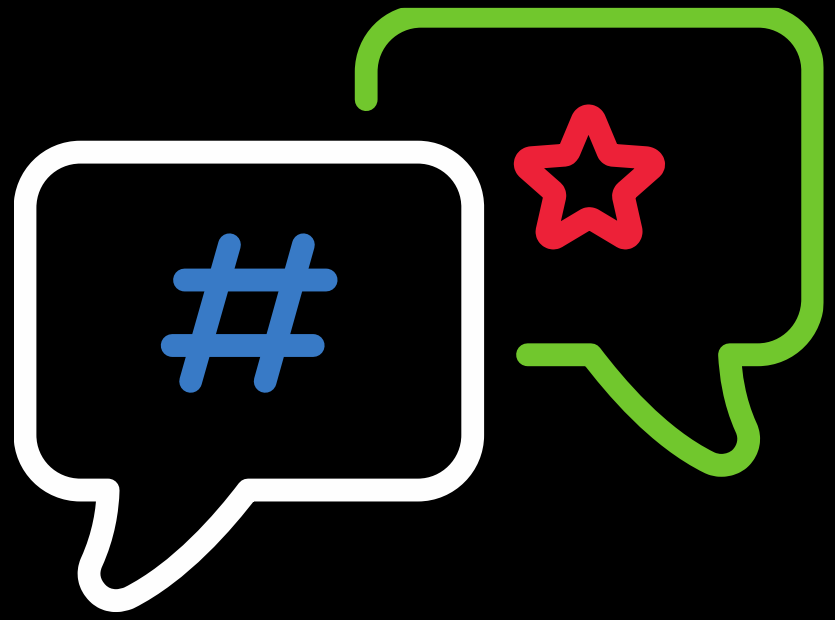
Let’s talk about: Data model and design considerations



Keep in mind

- **Open data model**: it is more a of creation than a configuration of a predefined tool.
- Many **alternative** paths of implementation can lead to the same solution. Evaluate **advantages and inconveniences** carefully.
- Whenever possible try to use an approach that verifies if the full solution works. Once the logic is assessed, the application can be refined. It allows us to identify all possible showstoppers sooner.
- Get frequent feedback and be open to review and change your data model design.

Let’s talk about: Data model concepts and design



Data model



Organize data

Determine data

Good data governance

Good insights

Data model

Dimensions (metadata)

- Axis of data analysis
- Elements to categorize data
- Quantify the business

Tables

- Dimensions
- Data (fact tables)
- Relationships
- Constraints

Data model design

The design is affected by

- Business rules (policies & procedures)
- Customer needs

Sources of information

- Managers, Users
- Reports, documentation

Examples of analysis outcomes

- Revenues collected by product
- Expense collected by cost centers
- Invoices by number, customer, address, product,...

Data modelling steps



1. Consider the **measures** of interest (Data)

2. Identify the **dimensions** that describe the measures

3. Isolate the dimensions with their **attributes** (Metadata)

4. Define the datasets. What is the structure of the fact **tables?**

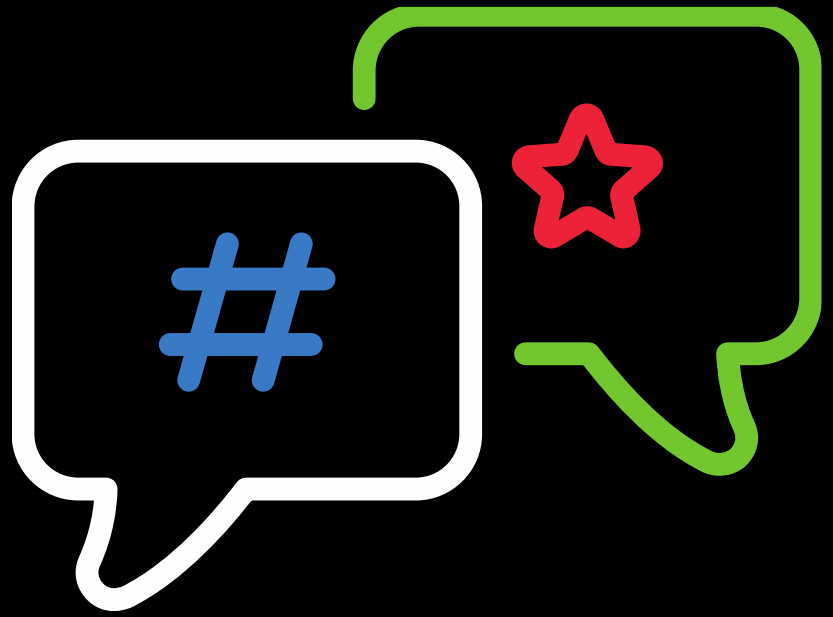
5. Establish the **relationships** between information in different tables.

Data model design – more aspects



- Split measures by **origin**:
  - calculated (and how)
  - loaded (and from)
  - manually inserted
- **Report** structure:
  - aggregation policies
  - filters
- Workflow and responsibilities
- User rights

Let’s talk about: Data model



Time dimension
Year-month vs Scenario-period

Year-month can be convenient for:

- calculations (they are numbers)
- to avoid process limitations

Scenario-period must always be present:

- for multidimensional reports
- to leverage standard functionalities (partitions, time rollup, conversions, navigations…)

Periods on columns can be convenient for:

- data entry (less rows)
- calculations

For any purpose, period dimension can be present and arbitrarily set (e.g. 12)

Account

It can be omitted also to avoid limitation in:

- process (editability checks)
- form (account is mandatory if present)

You can manage different logical accounts, based on data entry structure:

- using different measures
- on different columns

> ⚠ Pay attention to conflicts between account and measures in terms of rollup and conversion policies.

Entity



It drives the partitioning policy by business cycle & entity

It is used to generate calculation layer partition when DTP is run from context

With multidimensional process it is no more the only dimension driving the workflow

Currency

It represents the transactional currency

It is necessary to use the built-in conversion on-the-fly

More currencies can be managed with analytical dimensions if necessary

Other system dimensions: Category and Variation/Detail types

Their use is limited as in AIH there are not many built-in functionalities to leverage with these dimensions.

Use them to:

- leverage specific features available
- have a mapping with Financial workspace

Custom, analytical dimension or text field?



Custom dimensions

- Available in both FW and AW
- Classify financial data (what is the maximum level of financial detail?)
- Drive automatic mapping from AW to FW

Analytical dimensions

- Address specific needs in AW
- Drive reporting needs
- Can drive workflow
- User rights available

Text fields

- Allow definition of admitted values through dictionaries
- Free text needs no maintenance
- Filtering in forms is available

Dimension representing hierarchy nodes



In case of:

- Dimension A with many elements
- Use of aggregation in:
  - DTP calculation (allocations)
  - Dataset (data entry or virtual dataset)

It is recommended

- to manage aggregation on A through the use of another dimension B
- The elements of the B represent the hierarchy nodes of A (coding should be coherent)
- Association across dimensions is done through datasource reading from hierarchy of A

- Time dependency is guaranteed through data associations

Out of dictionary value allowed



By default, if a text field has a dictionary associated, it is possible to select only elements from the dictionary

For dimensions it is only possible to select dimensional elements

How to use a drop-down list and be able to specify exception, e.g. select all accounts starting with code 50…?

- Specify the field relative to account dimension as text field
- Associate a suitable dictionary
- Flag the “out of dictionary values allowed”
- Drop-down list is available, but it is possible to overwrite with any value

Process



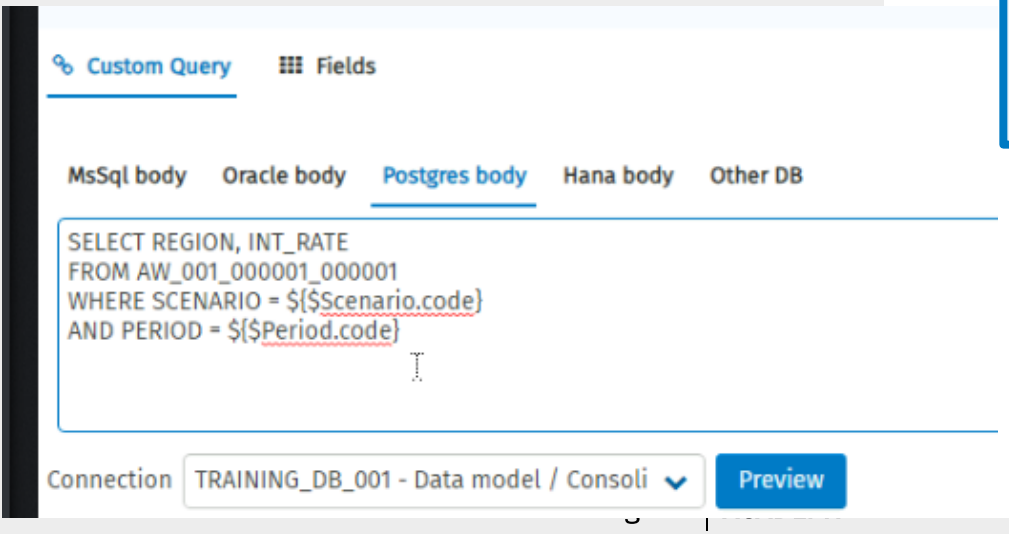
- Process data model configuration is used to drive editability
- If editability checks are not required, you can avoid process limitation if you don't use scenario, period, entity or account in datasets
- Process, especially multidimensional, is important, regardless of editability, to drive workflow
- Moving data from a process to another can be complex:
  - pull mode is preferable
  - datasource is the tool

Calculations

Datasets which are DTP target must have necessary fields for calculations.

Virtual datasets or datasources associated with DTP can be very useful to:

- Retrieve data from other workspaces
- Prepare data (preminary operations)
- Filter data



Datasets classification based on goals

Standard datasets

- Data entry
- Calculation
- Reporting

Reference datasets

- Configuration: definition of elements, rules, calculations...
- Transformation: look up tables...
- Assumptions: coefficients, data to be applied...

Datasources

- Retrieve metadata and data from other workspaces
- Prepare data for
  - calculations
  - reports

Virtual datasets

Prepare data for

- Calculations
- Reports
- Financial Workspace

Dataset

Use of multiple datasets or reduce them as much as possible?

There is not a general rule to state it. To take into consideration:

- Simplicity of structure
- Logical distinction among forms
- Measures sharing the same dimensionality
- Reporting needs
- Calculation needs
- Feeding sources
- High data volumes
- Partitioning

Dataset



IMPORTANT!

- Take time to assign proper and coherent **names** to dataset fields
- Fields with the same information should have the same label across datasets
  - Mapping of matching fields becomes easier
  - Reference dataset join and allocations use labels
- Base operations on codes, not on descriptions

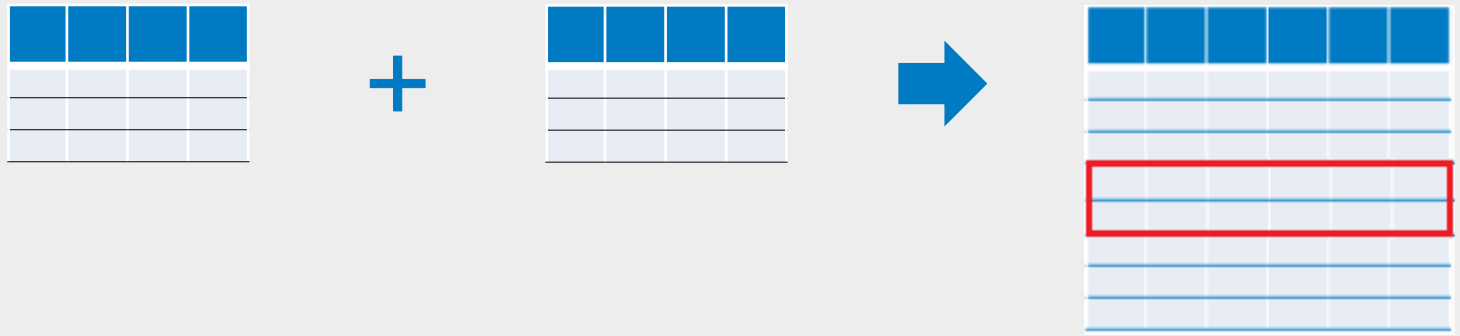
Naming – some advice

- **DIMENSION** for any dimensional type field (SCENARIO, PERIOD,…)
- **COD_XYZ** for any code defined as a text field.
- **DESC_XYZ** for any description
- **FLAG_XYZ** for any flag defined as a text field.
- **NUM_XYZ** for any numeric field related to non-financial measures.
- **DATE_XYZ** for any date field
- **AMOUNT_XYZ** for any numeric field related to financial measures.
- **AMOUNT** if there is only one amount field in the dataset.
- **TEXT##, NUM##** or **AMOUNT##** for multipurpose fields, e.g. fields used temporarily in DTP staging areas for calculations.

Best fit for Virtual dataset

Significant number of rows derived from smaller datasets

Data retrieve accesses a small portion of data involved (filters applied)



Best fit for Virtual dataset

Significant number of rows derived from smaller datasets

Data retrieve accesses a small portion of data involved (filters applied)

Source Dataset are refreshed quite frequently and/or the Virtual Dataset is accessed rarely

Avoid data processing task or data storage

Datasource vs Virtual dataset

Virtual dataset

- Can only derive data from datasets of the same AW
- Friendly definition
- Preferable to datasource when possible

Datasource

- SQL script, powerful but less friendly
- Use it for functions not managed and data not retrievable by virtual dataset

Datasource target

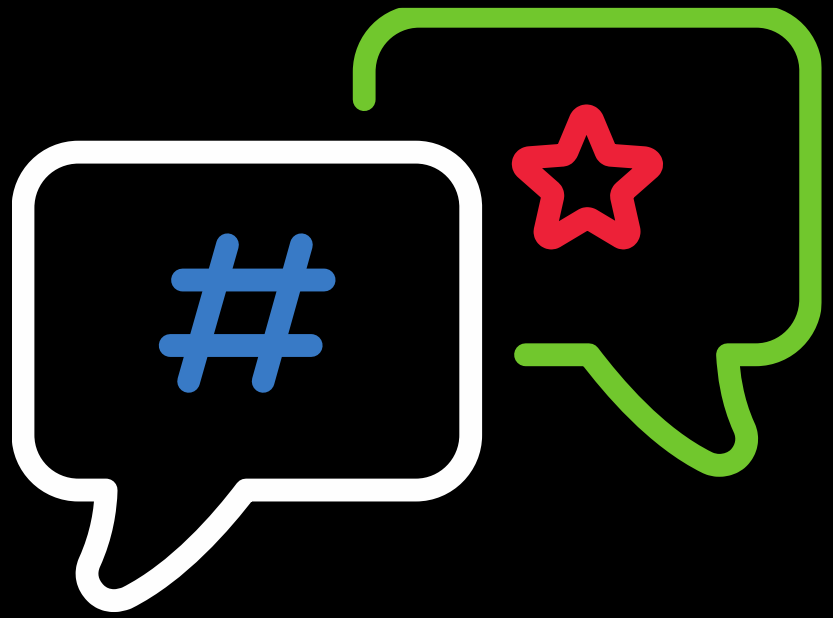
DTP

- Data preparation or extraction
  - Could be faster but less maintanable
- Operations not managed

Reporting

- Data not derivable from dataset
- Relationships and functions not managed by virtual dataset

Let’s talk about: Discussing solutions to specific needs



Case 1 Dataset enrichment



Total budgeted amount

PROJECT	COST CENTRE	ACCOUNT	AMOUNT
XXX	Division A	REVENUE	300000
YYY	Division B	DIRECT COST	200000
ZZZ	Division C	OTHER COST	50000

Invoices

PROJECT	SUPPLIER	INVOICE DATE	ACCOUNT	AMOUNT
XXX	Supplier 1	03-03-2020	DIRECT COST	10000
XXX	Supplier 2	21-04-2020	OTHER COST	5000
YYY	Supplier 1	25-04-2020	DIRECT COST	8000

Case 1 Dataset enrichment



Goal

Calculate remaining amount by project, cost centre, account and make it available for further calculations

PROJECT	COST CENTRE	ACCOUNT	BUDGET AMOUNT	INVOICED AMOUNT	REMAINING AMOUNT
XXX	Division A	REVENUE	300000	30000	270000
YYY	Division B	DIRECT COST	200000	10000	190000
ZZZ	Division C	OTHER COST	50000	8000	42000

Case 1 Dataset enrichment



Which solution to use?

- Excel formula
- DTP writing on a new Dataset
- Virtual Dataset
- DTP writing on one of existing Datasets

Case 1 Dataset enrichment



Analysis of solutions

Excel formula

- Not easy or convenient to use excel formula across different dataset (e.g. difficulties to join columns)

Virtual dataset

- No grouping
- Necessity of a DTP

DTP writing on one of existing dataset

- Shared dimensionality, less dataset
- Necessary to add two columns
- Dataset fed by two different sources: possible conflicts

DTP writing on a new dataset

- Better tracking of data flow

Case 2
Overwrite prepopulation



Definition

ASSET CLASS	PERIODS OF USERFUL LIFE
Machinery	50
Equipment	20
Building	120
…	…

Prepopulation

ASSET CLASS	PERIOD	DEPRECIATION
Machinery	1	2%
Machinery	2	2%
Machinery	3	2%
Machinery	4	2%
…	…	…
Equipment	1	5%
Equipment	2	5%
Equipment	3	5%
…	…	…

Case 2
Overwrite prepopulation



Goal

Manual overwrite

ASSET CLASS	PERIOD	DEPRECIATION
Machinery	1	2%
Machinery	2	2%
Machinery	3	2%
Machinery	4	2%
…	…	…
Equipment	1	10%
Equipment	2	10%
Equipment	3	5%
…	…	…

Avoid that running prepopulation deletes the manually overwritten percentages

Case 2
Overwrite prepopulation



Which solution to use?

ETL filters

Specific DTP operations

Reporting layer filter in DTP

Case 2 Overwrite prepopulation



Analysis of solutions

ETL filters

- Easy solution with deletion and insertion filters
- DTP is specific for AW and it should be used when possible

Reporting layer filters in DTP

- Deletion filter
  - It can avoid the cancellation of the amount overwritten
  - It cannot avoid the insertion of the prepopulation

Specific DTP operation

- Enrichment from another dataset that checks if an overwritten percentage is existing

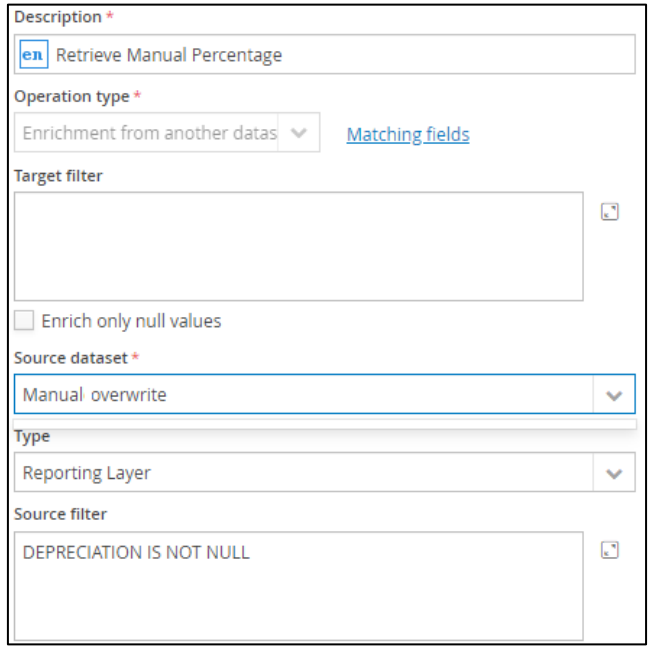
Case 2 Overwrite prepopulation



Solution

Manual overwrite

ASSET CLASS	PERIOD	DEPRECIATION
Machinery	1	2%
Machinery	2	2%
Machinery	3	2%
Machinery	4	2%
…	…	…
Equipment	1	10%
Equipment	2	10%
Equipment	3	5%
…	…	…



Case 2
Overwrite prepopulation



Solution

Manual overwrite

ASSET CLASS	PERIOD	DEPRECIATION
Machinery	1	2%
Machinery	2	2%
Machinery	3	2%
Machinery	4	2%
…	…	…
Equipment	1	10%
Equipment	2	10%
Equipment	3	5%
…	…	…

ASSET CLASS	Condition
DEPRECIATION	Attribute
PERIOD	Condition

Case 3
Unpivot data



Dataset with 12 amount fields representing different periods

Dimensions…	**P01**	P02	P03	…	**P11**	P12
XXX	**10000**	3000	6000	…	**6000**	2000
XZY	**5000**	7000	7000	…	**3000**	2000
ZXY	**8000**	5000	3000	…	**1000**	8000

Case 3
Unpivot data



Goal

Unpivot data from the structure with 12 amount columns to a structure with one column for periods and one for amount

Dimensions	PERIOD	AMOUNT
XXX	Jan	1000
XXX	Feb	3000
XXX	Mar	6000
XXX	Nov	6000
XXX	Dec	2000
…	…	…
XZY	Jan	5000
XZY	Feb	7000
…	…	…

Case 3
Unpivot data



Which solution to use?

- Virtual dataset (+ DTP)
- Datasource (+ DTP)
- Special method
- FW mapping

Case 3
Unpivot data



Analysis of solutions

FW mapping

- Native functionality to unpivot data
- Only if all the dimensions are also present in FW

Virtual dataset

- User friendly
- Necessity of a reference dataset to open rows
- DTP to finalize and store results

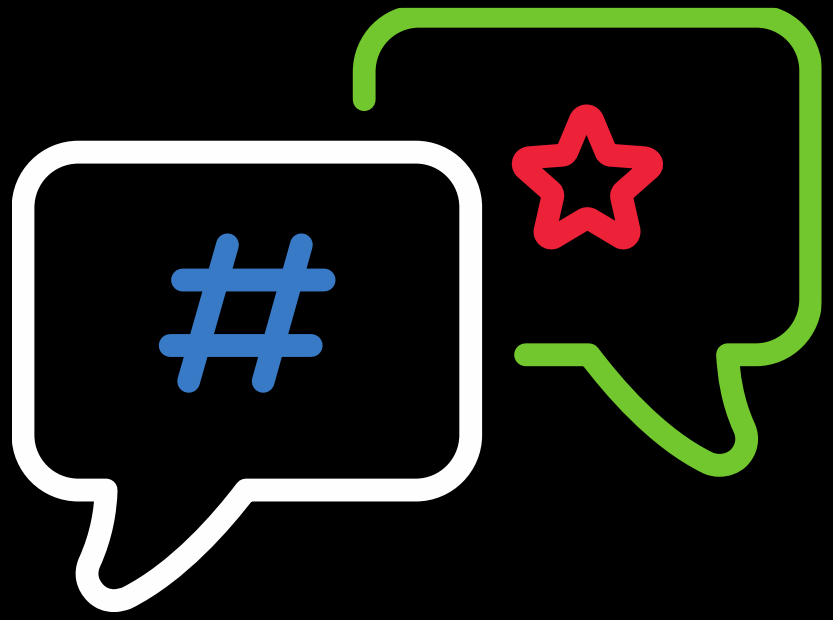
Datasource

- SQL script (with different level of complexity)
- DTP if necessary

Special method

- Datasource has the same funcionality and it is preferable

Let’s talk about: Business case



Exercises theory



- Hands on Goals & Data Model
- Creation of an AW Dictionary
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- Replace old parameters with new ones
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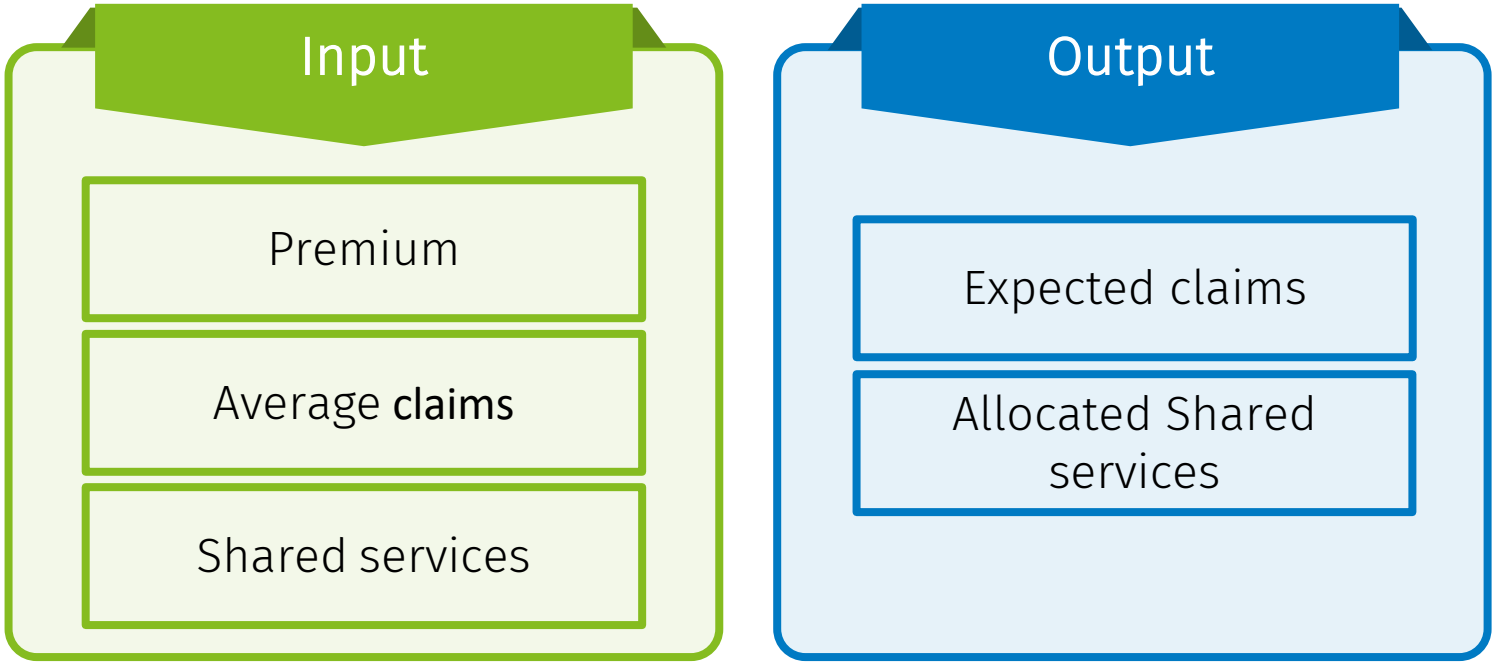
Goal

Expected claims

Shared services allocated

	2050 Forecast
Premium	435,730
Expected claims	240,000
Loss Ratio	**55.1%**
IT Allocated	32,394
General services allocated	36,443
Marketing & Sales Allocated	28,345
Expense Ratio	**22.3%**

Input and output

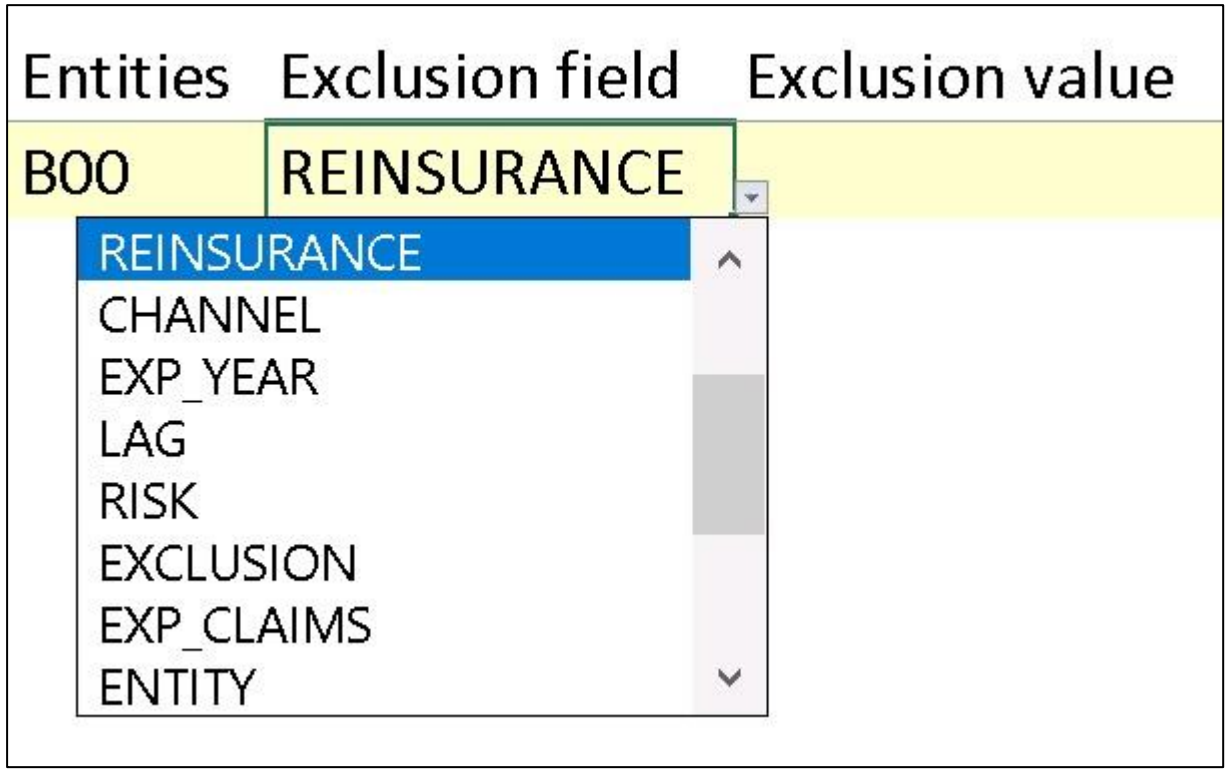


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Creation and use of an AW Dictionary



Tables

- Dimension
  - List of elements: ED_[Analytical dimension label]
  - Hierarchy structure: ED_[label]_GERARCHIA
  - Hierarchy relationships: ED_[label]_GERARCHIA_ABBI
- List of Analytical Workspace: OWS (key is the ID, not the code)
- List of Dataset: OWS_TABELLA
- List of all dataset fields (columns): OWS_TABELLA_CAMPO
- List of physical table of the partitions: OWS_TABELLA_PARTIZIONE

Tables

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- List of physical table of the partitions: OWS_TABELLA_PARTIZIONE

Retrieving ID from Dataset URL



- In a Dataset's definition URL:
  - String after last «/» represent the name of ID_TABELLA into AW Tables

Fact tables naming

AW

Code AW

Dataset

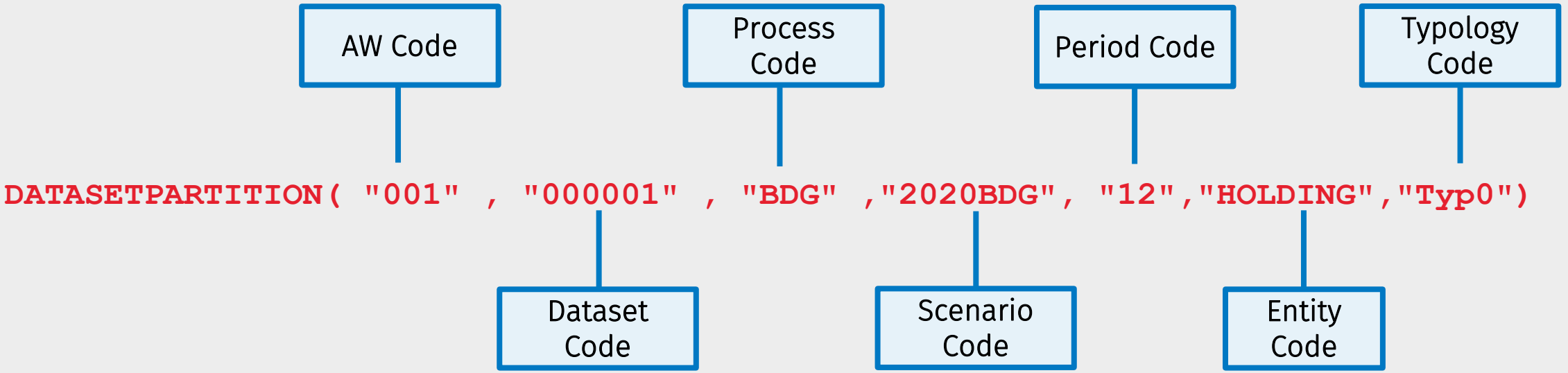
Partition

AW_001_000001_000001

Physical name of a dataset partition

Cell field - DATASETPARTITION ( ):

in drill-through queries, DTP, Datasource mapping, SQL clause retrieve the name of dataset partition based on its names and context



PROCESS($Scenario.code,$Period.code).code → Useful to retrieve Process code

Dependency between fields

- Dependent drop-down lists are created recalling in two dataset fields two dictionary fields associated
- When first field is selected, the list relative to second field is automatically filtered
- Data related to fields dependency are stored in a dataset
- Create a dictionary to retrieve values from dependency dataset
- Set properly «Field Value» And «Query field code» in the related fields of destination dataset to create dependency with the dictionary

Now it’s your turn

Exercise # 1



Exercises theory



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Feeding dataset fields from virtual dataset

- Create a virtual dataset
- In fields tab define a field whose code is equal to one field of the joined dataset
- Match the two fields Into a DTP operation (e.g. Insert from another dataset)
- Phisycal dataset field will be fed by Virtual dataset field

Original dataset Premium

Entity	Region	....	Reinsurance	Premium	Exclusion formula	Exclusion flag
B00	ITN	.....	A	1000		
B00	ITC	.....	C	-2000		
B00	ITS	.....	C	3000		

[40 Rows]

Fill exclusion formula by virtual dataset

Entity	Region	....	Reinsurance	Premium	Exclusion formula
B00	ITN	.....	A	1000	REINSURANCE =C
B00	ITN	.....	A	1000	PREMIUM <0
B00	ITC	.....	C	-2000	REINSURANCE=C
B00	ITC		C	-2000	PREMIUM <0
B00	ITS		C	3000	REINSURANCE =C
B00	ITS		C	3000	PREMIUM <0

[80 Rows]

Valued by virtual dataset

Total rows are doubled due to the 2 exclusion formula

Fill Exclusion flag by Enrichment with formula field

Entity	Region	....	Reinsurance	Premium	Exclusion formula	Exclusion flag
B00	ITN	.....	A	1000	REINSURANCE =C	0
B00	ITN		A	1000	PREMIUM <0	0
B00	ITC	.....	C	-2000	REINSURANCE =C	1
B00	ITC		C	-2000	PREMIUM <0	1
B00	ITS	.....	C	3000	REINSURANCE =C	1
B00	ITS		C	3000	PREMIUM <0	0

Valued by Enrichment with formula field

[80 Rows]

Now it’s your turn

Exercise # 2



Exercises theory



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Grouping: summing up the exclusion flag

Entity	Region	....	Reinsurance	Premium	Exclusion formula	Exclusion flag
B00	ITN	.....	A	1000		0
B00	ITC	.....	C	-2000		2
B00	ITS	.....	C	3000		1

[40 Rows]

User Exclusion will be performed selecting items with **Exclusion field = 0**

Now it’s your turn

Exercise # 3



Exercises theory



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Filtering methods

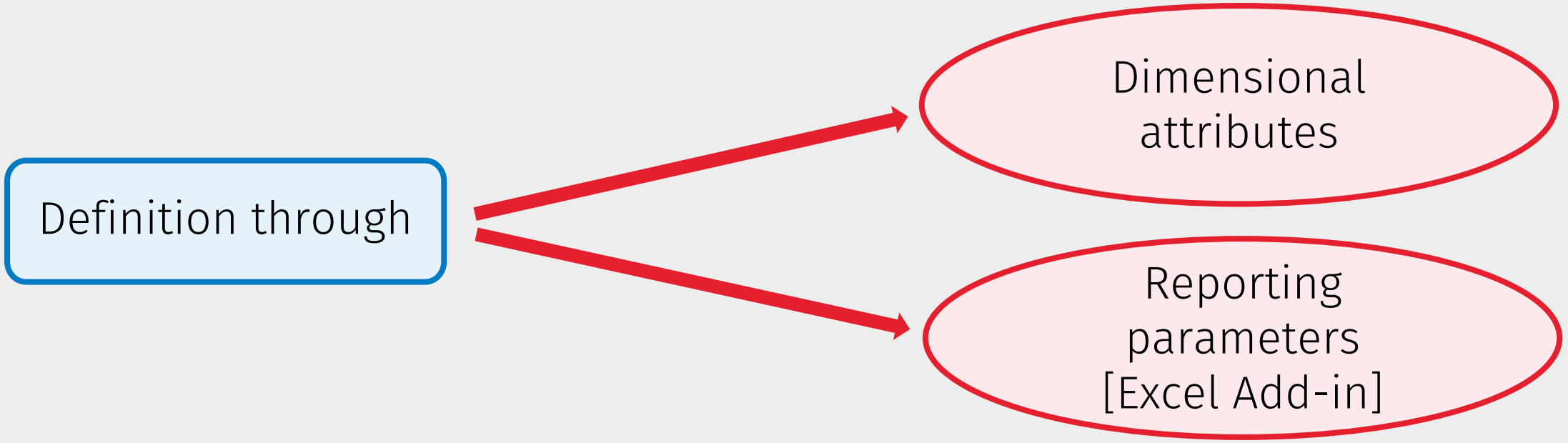
- Origin condition field
  - Basic SQL skills
  - Low level of customization
- Special method
  - SQL skills
  - Maintenance needed
  - Does what you type not what you want!
  - Reusable in more DTP of same AW

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New types of parameters: definition



Reporting parameters selection from

- Dictionary
- Date
- Text
- Number
- Flag
- Dimension

New types of parameters: where to use

Usable in

- Any DTP/SQL clause
  - Filters
  - Run conditions
  - Enrichment with formulas
  - Database Objects
  - Special methods
  - Datasources

Not available with dictionaries

Parameters recalling

Dimensional attribute

- ${$Entity.code}

Reporting parameter

- ${@PARAMETER}

Syntax examples

${$Entity.currency.code}

- Dimensional attribute retrieving entity currency

${@FLAG}='1'

- Flag parameter identity used in Run condition

${@DICTIONARY}

- Reporting parameter enriching with elements selected from dictionary

${@$ANL_REGION_10_P.parent.code}

- Reporting parameter mixed with dimensional attribute retrieving the parent of an element selected from an analytical dimension hierarchy

Exercises theory



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Old parameters	Old Syntax	How to replace
Workspace @...@	@PARAM@	Reporting Parameter
System #...#	#ENTITIES#	Attribute: ${$Entity.code}
	#SCENARIOS#	Attribute: ${$Scenario.code}
	#PERIODS#	Attribute: ${$Period.code}
	#TARGET_PARTITION#	Attribute: ${DATASETPARTITION(....)}
Special Method &...&	&TABLE1&	Attribute: ${DATASETPARTITION(....)}
	&STRING1&	Reporting parameter: Text

Enrichment of Premium's missing fields

Lag is the difference in months between running Scenario/Period and expiring date

Probability Factor

IELR* Avg Claims

Exp year	Exp month	Region	Lag	IELR	Avg Claims	Exp Claims
2050	12	ITN			2000	
2051	02	ITC			3000	
2050	10	ITS			4000	

Now it’s your turn

Exercise # 4



Exercises theory



- Hands on Goals & Data Model
- Creation of an AW Dictionary
- Virtual dataset advanced use
- Grouping function
- Filtering methods
- Parameters
- Replace old parameters with new ones
- Levels in reference dataset
- Insert with reference dataset
- Reporting with custom filters
- Allocations
- Profit & Loss reporting

Insert with reference dataset

Insert with reference dataset

1) Insert from another dataset
2) Enrichment with Reference dataset
3) Enrichment with formula

→ 3 Operations in 1

⚠
- Supports all filters and all Reference dataset types
- Level field on the reference table does not work in this operation

Solution without Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000
Software	United Kingdom	20000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21
United Kingdom	0.20

Target

Product	Region	Sales	VAT	Account receivable

Solution without Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000
Software	United Kingdom	20000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21
United Kingdom	0.20

Target

Product	Region	Sales	VAT	Account receivable
Software	Italy	30000		
Hardware	Spain	10000		
Software	United Kingdom	20000		

Acc. Receivable - Operations

- Insert from another dataset

Solution without Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000
Software	United Kingdom	20000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21
United Kingdom	0.20

Acc. Receivable - Operations

- Insert from another dataset
- Enrichment with ref. dataset

Target

Product	Region	Sales	VAT	Account receivable
Software	Italy	30000	0.22	
Hardware	Spain	10000	0.21	
Software	United Kingdom	20000	0.20	

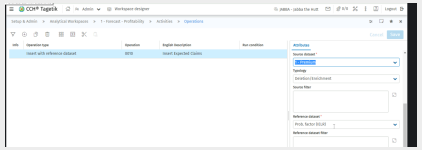
Solution without Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000
Software	United Kingdom	20000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21
United Kingdom	0.20



Acc. Receivable - Operations

- Insert from another dataset
- Enrichment with ref. dataset
- **Enrichment with formula**

Account Receivable =
Sales X (1+VAT)

Target

Product	Region	Sales	VAT	Account receivable
Software	Italy	30000	0.22	36 600
Hardware	Spain	10000	0.21	12 100
Software	United Kingdom	20000	0.20	24 000

Solution: Insert with reference dataset



Insert with reference dataset needs only to setup attributes

Target

Product	Region	Sales	VAT	Account receivable

Solution: Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000



Insert with reference dataset needs only to setup attributes

Target

Product	Region	Sales	VAT	Account receivable

Solution: Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21



Insert with reference dataset needs only to setup attributes

Target

Product	Region	Sales	VAT	Account receivable

Solution: Insert with reference dataset

Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21

Matching fields

Target field	Source
Account receivable	**SOURCE**.Sales * (1 + **REF**.VAT)

Target

Product	Region	Sales	VAT	Account receivable



Insert with reference dataset needs only to setup attributes

Solution: Insert with reference dataset

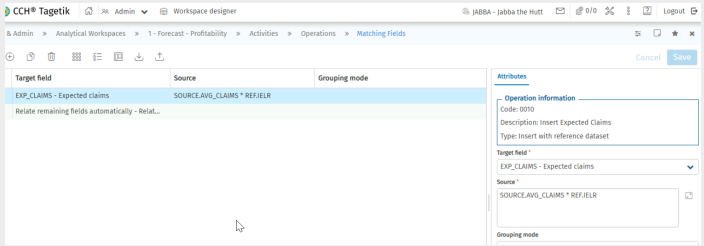
Source

Product	Region	Sales
Software	Italy	30000
Hardware	Spain	10000

> Insert with reference dataset needs only to setup attributes

Reference Dataset

Region	VAT
Italy	0.22
Spain	0.21



Matching fields

Target field	Source
Account receivable	**SOURCE**.Sales * (1 + **REF**.VAT)

Target

Product	Region	Sales	VAT	Account receivable
Software	Italy	30000	0.22	36 600
Hardware	Spain	10000	0.21	12 100

Exercises theory



- Hands on Goals & Data Model
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- Levels in reference dataset
- Insert with reference dataset
- Reporting with custom filters
- Allocations
- Profit & Loss reporting

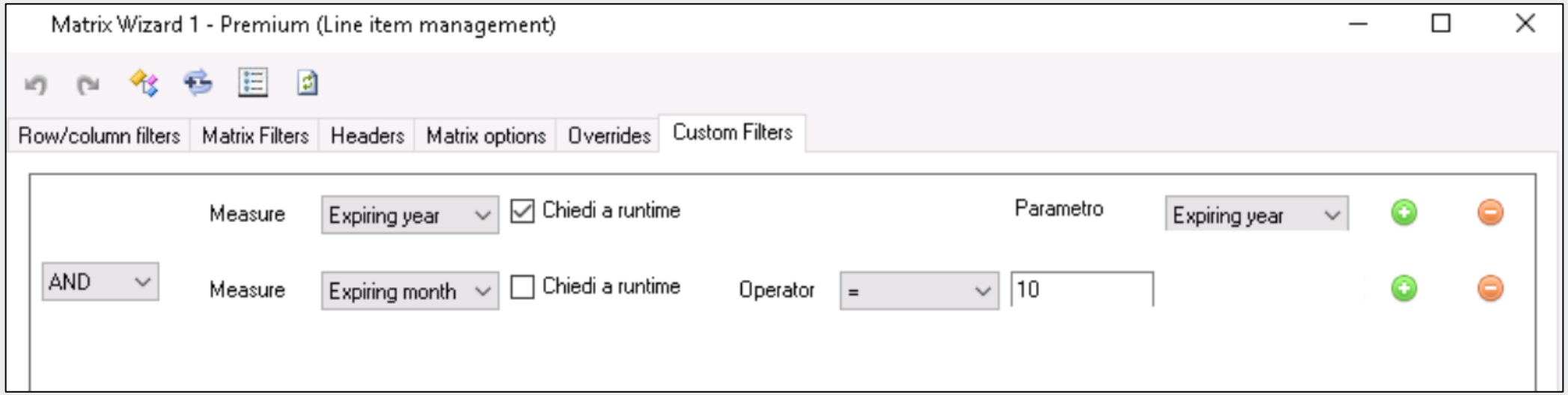
Custom filters in reporting

Applicability

- Line-by-line matrices
- Measures: text, date and number

Filter specification

- Hardcoded value
- Asked at runtime through parameter



Now it’s your turn

Exercise # 5



Exercises theory

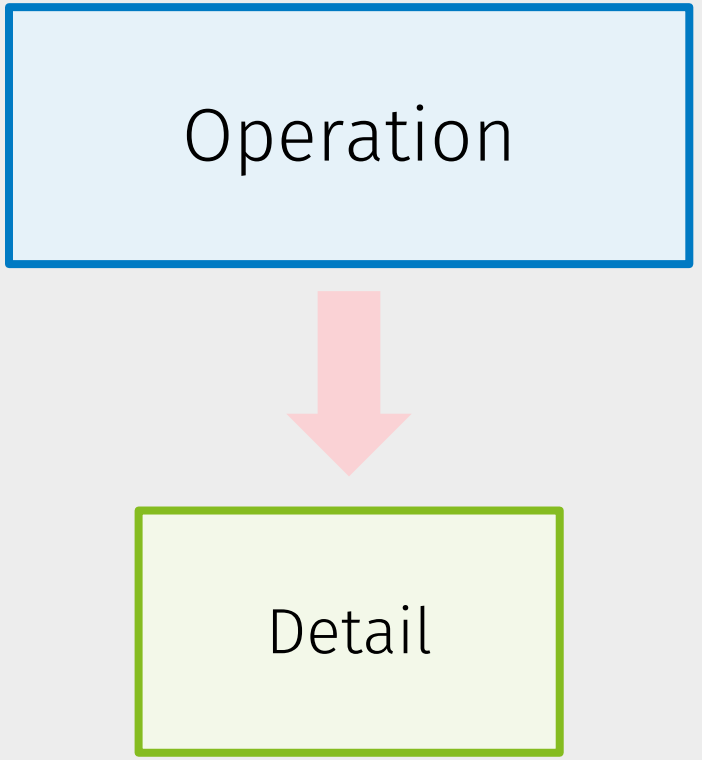


- Hands on Goals & Data Model
- Creation of an AW Dictionary
- Virtual dataset advanced use
- Grouping function
- Filtering methods
- Parameters
- Replace old parameters with new ones
- Levels in reference dataset
- Insert with reference dataset
- Reporting with custom filters
- Allocations
- Profit & Loss reporting

Allocation operation structure

- OPERATION
  - Driver dataset
  - Dataset data to be allocated

- DETAIL
  - Measure to allocate
  - Driver measure
  - Dimensions on which allocate



Allocation usage restrictions



Matching field not present: Dataset to be allocated and Target dataset must have the fields coded in the same way

Not suitable for allocating more measures in a single operation

Not suitable for allocating on Scenario, Period, Entity

Exercises theory



- Hands on Goals & Data Model
- Creation of an AW Dictionary
- Virtual dataset advanced use
- Grouping function
- Filtering methods
- Parameters
- Replace old parameters with new ones
- Levels in reference dataset
- Insert with reference dataset
- Reporting with custom filters
- Allocations
- Allocation of multiple measures in the same operation
- Profit & Loss reporting

Dataset Profit & Loss: allocation rows

Channel	LOB	Risk	Region	Entity	Scenario	Period	Cost Center	Account	Amount
							CC01 – IT	00030	200
							CC02 – MK	00030	350
							CC03 - GS	00030	110

Dimensions on which allocate

Inherited by the environment

Source data

Allocation calculation

Goal

Create this multidimentional matrix from dataset Profit & Loss

	2050 Forecast
Premium	435,730
Expected claims	240,000
Loss Ratio	**55.1%**
IT Allocated	32,394
General services allocated	36,443
Marketing & Sales Allocated	28,345
Expense Ratio	**22.3%**



Premium and Expected Claims should be loaded from dataset Premium

Dataset Profit & Loss: insert premium & claims

Use Insert from another dataset

Channel	LOB	Risk	Region	Entity	Scenario	Period	Cost Center	Account	Amount
							CC01 – IT	00030	200
							CC02 – MK	00030	350
							CC03_GS	00030	110
							GEN_CC	00010	5000

Automatic matching

Fed by cell field
$Scenario.code
$Period.code

Cost Center
Not present in
Premium

Specific
account

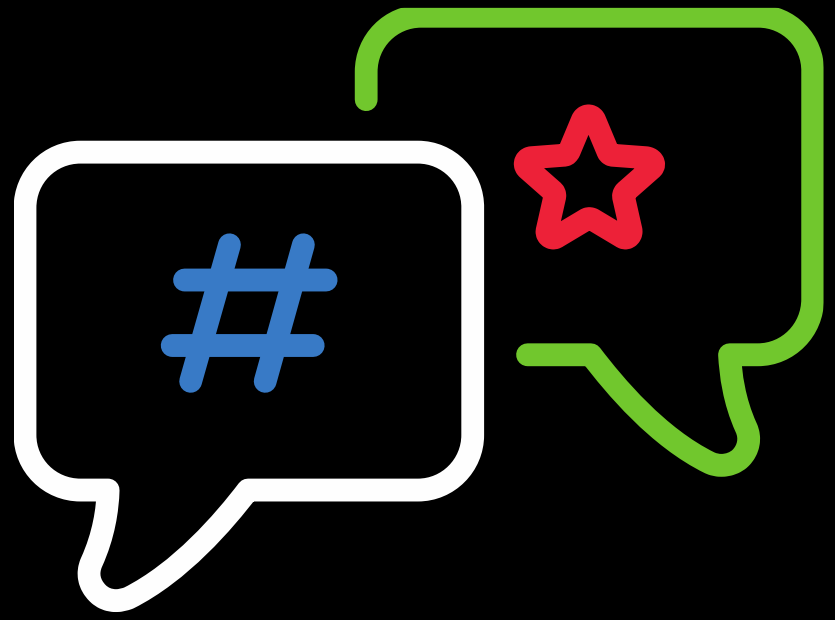
Premium
measure

Now it’s your turn

Exercise # 6



Let’s talk about: Technical guidelines

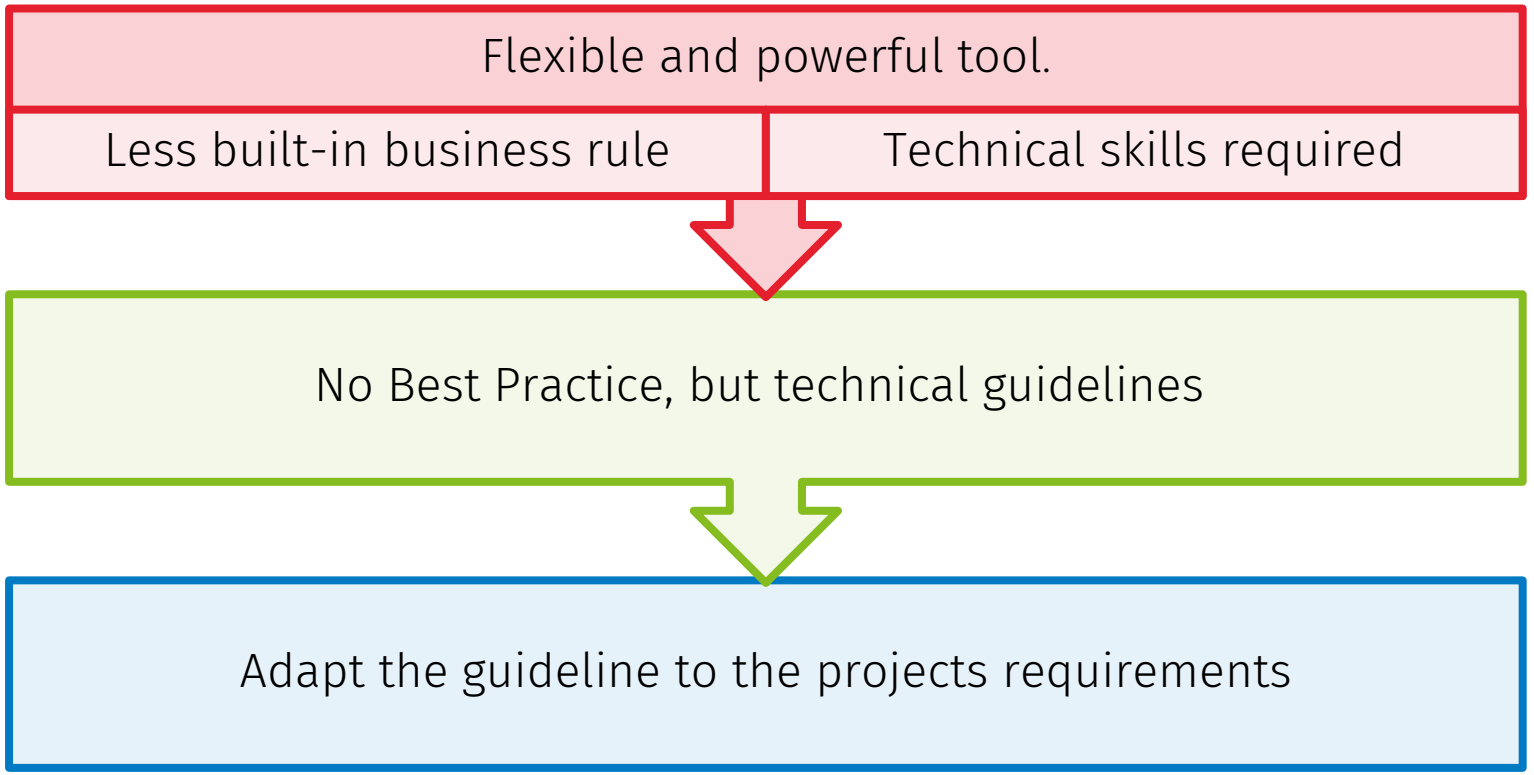


Technical guidelines



- Why the technical perspective is important
- Dataset partitioning
- Dataset indexing
- DTP
- Special Method

Why technical perspective?

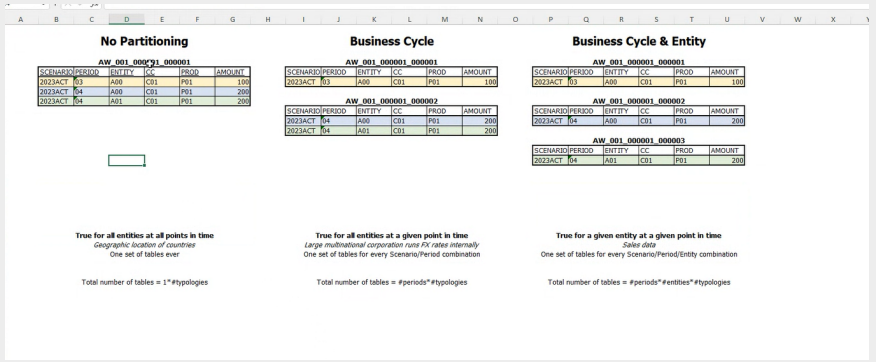


Technical guidelines



- Why the technical perspective is important
- Dataset partitioning
- Dataset indexing
- DTP
- Special Method

How should dataset be partitioned?



No Partitioning

AW_001_000001_000001

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	03	A00	C01	P01	100
2023ACT	04	A00	C01	P01	200
2023ACT	04	A01	C01	P01	200

True for all entities at all points in time
Geographic location of countries
One set of tables ever

Total number of tables = 1*#typologies

Business Cycle

AW_001_000001_000001

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	03	A00	C01	P01	100

AW_001_000001_000002

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	04	A00	C01	P01	200
2023ACT	04	A01	C01	P01	200

True for all entities at a given point in time
Large multinational corporation runs FX rates internally
One set of tables for every Scenario/Period combination

Total number of tables = #periods*#typologies

Business Cycle & Entity

AW_001_000001_000001

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	03	A00	C01	P01	100

AW_001_000001_000002

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	04	A00	C01	P01	200

AW_001_000001_000003

SCENARIO	PERIOD	ENTITY	CC	PROD	AMOUNT
2023ACT	04	A01	C01	P01	200

True for a given entity at a given point in time
Sales data
One set of tables for every Scenario/Period/Entity combination

Total number of tables = #periods*#entities*#typologies

None

- Low data volumes
- Roster, list of elements
- Attributes not Scenario/Period dependent

By Business Cycle

- Native segmentation of historical data (volumes do not increase moving on time)

By Business Cycle and Entity

- Management of data by project – There is not an actual business cycle, but a full life project management

Dataset partitioning: seeking the right target

The partition contains all and only data needed during each single request

Partition contains only data of a single actual business cycle

Concurrent activities target distinct partitions

Partitions and process-scenario management

To leverage as much as possible partitioning benefits, avoid switching scenarios within the same process: every business cycle has to be managed through his own process

Partitions and process-scenario management

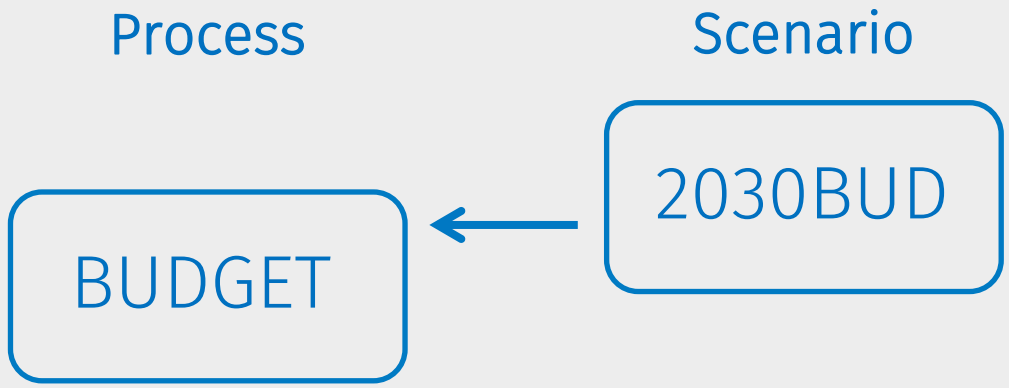
To leverage as much as possible partitioning benefits, avoid switching scenarios within the same process: every business cycle has to be managed through his own process

Process

BUDGET

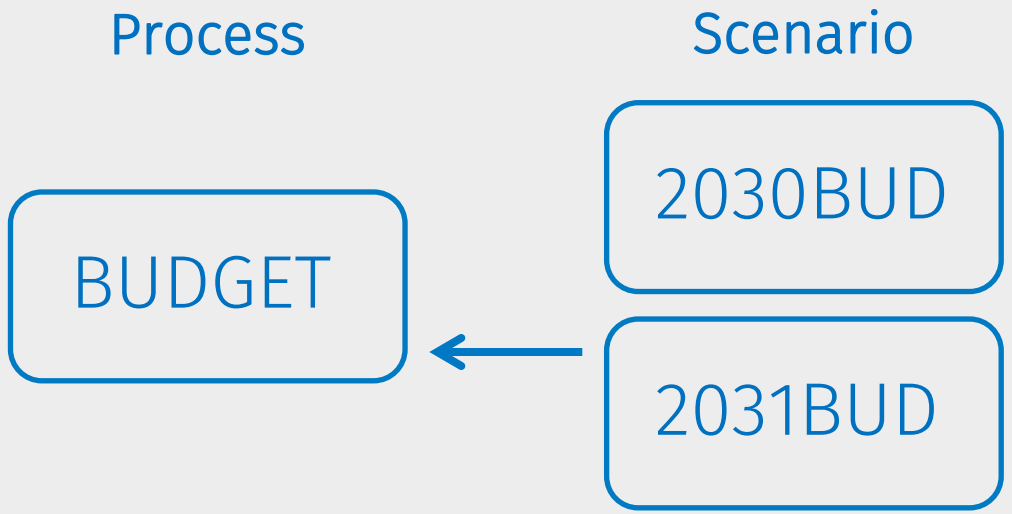
Partitions and process-scenario management

To leverage as much as possible partitioning benefits, avoid switching scenarios within the same process: every business cycle has to be managed through his own process



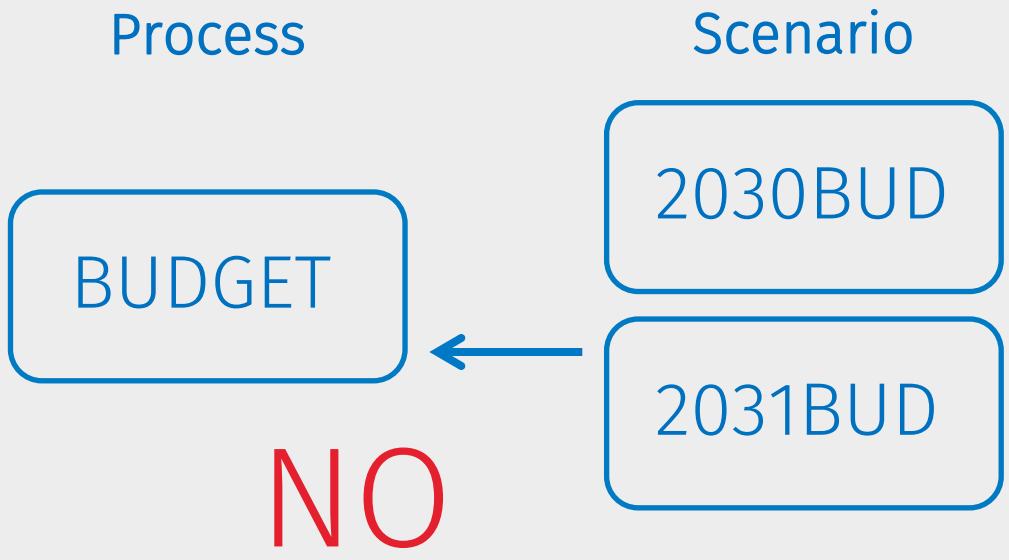
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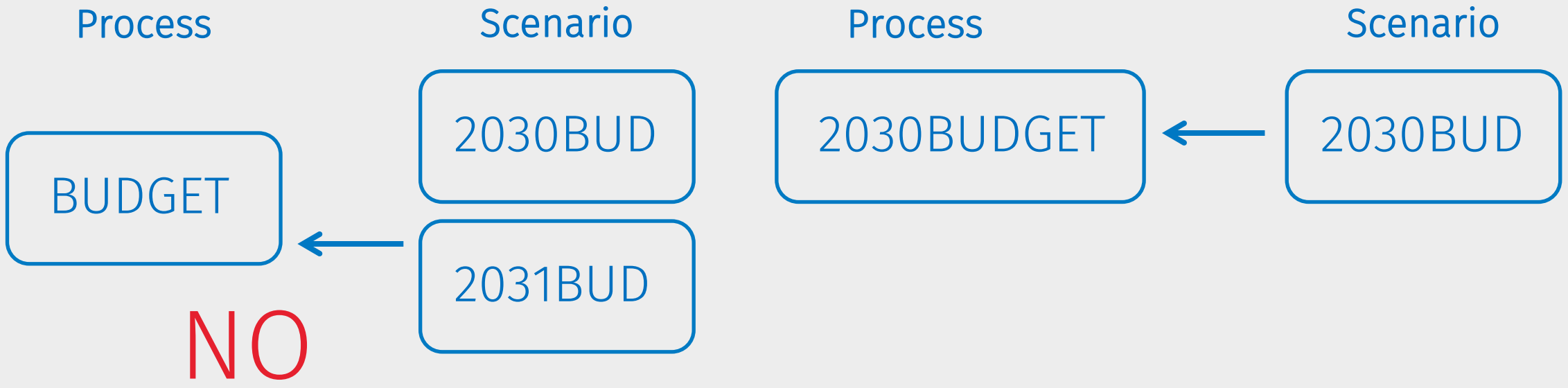
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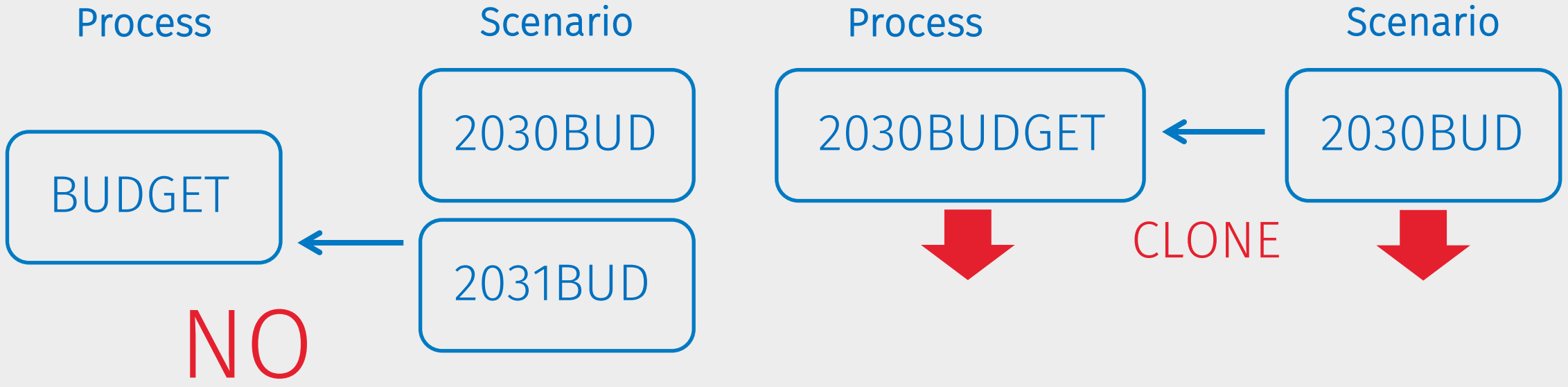
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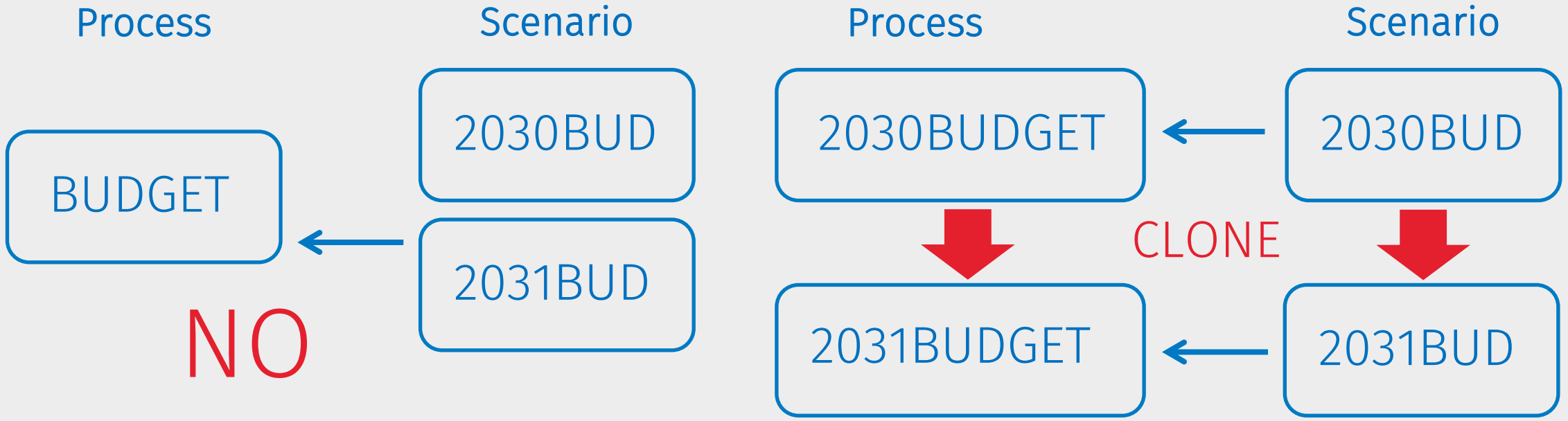
Partitions and process-scenario management

To leverage as much as possible partitioning benefits, avoid switching scenarios within the same process: every business cycle has to be managed through his own process



Partitions and process-scenario management

To leverage as much as possible partitioning benefits, avoid switching scenarios within the same process: every business cycle has to be managed through his own process



Technical guidelines



- Why the technical perspective is important
- Dataset partitioning
- Dataset indexing
- DTP
- Special Method

Dataset indexing: basic principles

- What is a database index ?
  - It is a structure, physically stored, that improves the speed of data retrieval operations
- What does a dataset index mean?
  - For each Dataset, an indexing criteria can be defined
  - An index is then created for each partition of the dataset

Dataset Indexing

Dataset – Index on CC

Row	Cost Centre	Other fields	...
1	CC1	...	...
2	CC1		
3	CC2		
4	CC1		
5	CC3		
6	CC2		
7	CC1		

Dataset Indexing

Index table

Element	Row
CC1	1,2,4,7
CC2	3,6
CC3	5

Dataset – Index on CC

Row	Cost Centre	Other fields	…
1	CC1	…	…
2	CC1		
3	CC2		
4	CC1		
5	CC3		
6	CC2		
7	CC1		

Dataset Indexing

Index table

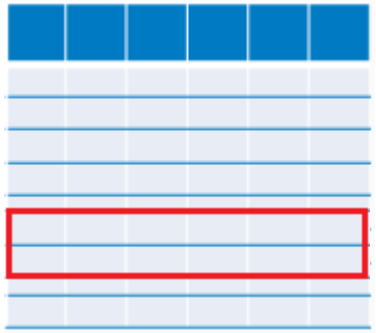
Element	Row
CC1	1,2,4,7
CC2	3,6
CC3	5

Dataset – Index on CC

Row	Cost Centre	Other fields	…
1	CC1	…	…
2	CC1		
3	CC2		
4	CC1		
5	CC3		
6	CC2		
7	CC1		

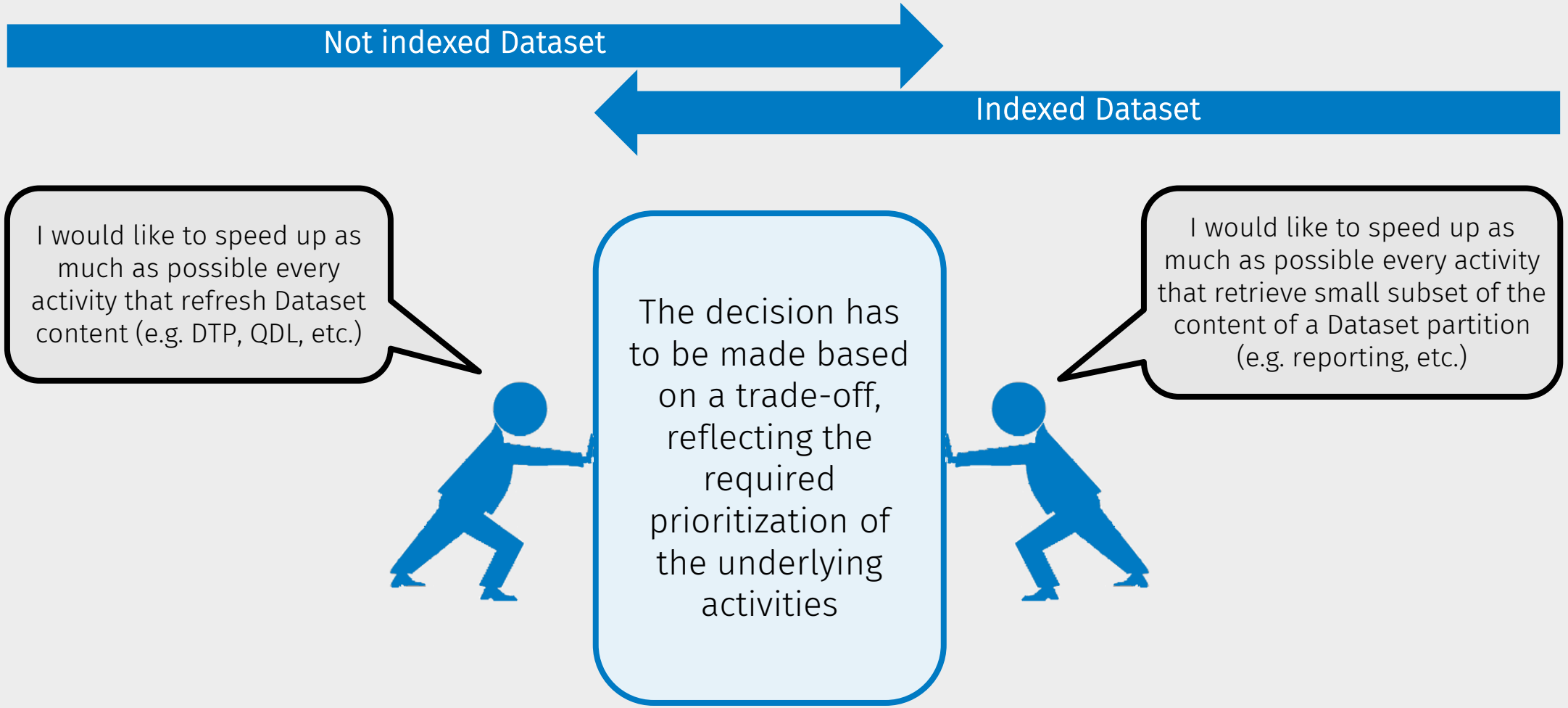
Dataset indexing: features

- additional selective criteria with respect to the partitioning
- mechanism aimed to retrieve data quickly
- negatively impact data update
- effective only when accessing a reduced part of the partition



< 20% of the full table

Dataset indexing: finding the best trade-off



Technical guidelines



- Why the technical perspective is important
- Dataset partitioning
- Dataset indexing
- DTP
- Special Method

DTP execution

Standard

- Default
- Execution by entity and by node
- Data processed and stored in calculation layer partitions
- Not available in multidimensional process

Pro

- Staging area is accessible
- Debug

Context independent run

- Enabled in workflow - DTP task definition
- Parallel execution by user
- Data processed on temporary partitions and not stored
- The only mode available for multidimensional processes

Pro

- More performance, especially in project with high levels of concurrency
- Calculation layer space is saved

DTP on HANA



- Enrichment/join from another dataset
- Enrichment with formula

→

Run row by row

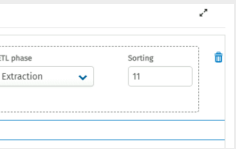
→

More efficient only on HANA

FOUND_ADVANCED - Advanced feature... 0003874 - ETL - Commands Admin All Modules

Apply

Module	Function	License type	Edit type
FOUND_ADVANCED - Advanced feature...	0002305 - Query Analyzer	Admin All Modules	All



Pros

- Extremely flexible and powerful
- Extend capability beyond existing built-in functionalitied

Cons

- **Advanced knowledge of SQL** language is required
- **Deep technical knowledge** of CCH Tagetik structure
- **Compatibility not guaranteed after CCH Tagetik upgrade**
- **No log nor auditing trail**
- The are no constraints on the executed code. "**User Rights**" concept does not apply
- New technologies not leveraged

Module	Function	License type	Edit type
FOUND_ADVANCED - Advanced feature...	0002306 - Query Analyzer (Restricted)	Admin All Modules	Don't modify data



Takeaway

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Share the love by participating in Testimonials, Case studies, Surveys, Reviews, Referrals

Leverage your social capital by connecting with other finance and operations professionals

Earn points up the leader board and redeem rewards

Here's the way the hub works

1 Log into the CCH Tagetik Community and click on the Insiders tab

2 Poke around and complete challenges earning points to redeem rewards

3 Get recognized for all the support you give CCH Tagetik

It's that easy!

Join CCH Tagetik Insiders now!

What other learning assets are available?

User Manual

CCH Tagetik Community

Knowledge Base

What's next?

Visit the Academy webpage

Next training and/or Assessment

Mail to info.academy@wolterskluwer.com

Thank you



Wolters Kluwer

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Appendix



- Allocation of multiple measures in the same operation (optional)

Allocation of more measure of a dataset

It is necessary create an operation for each measure

Each detail of an operation runs as it is alone not considering the other ones

All dataset fields not involved in allocation are copied as they are

Which is the result?

Allocation of more measure - Result

What I expect

Period	Scenario	Channel	Marketing & Sales Allocated	IT Costs allocated	GS Costs allocated
06	2050FCT	CH_010	**3,000**	**4,500**	**9,000**
06	2050FCT	CH_015	**2,000**	**3,000**	**6,000**
06	2050FCT	CH_020	**5,000**	**7,500**	**15,000**

What I get

Period	Scenario	Channel	Marketing & Sales Allocated	IT Costs allocated	GS Costs allocated
06	2050FCT	CH_010	10,000	15,000	**9,000**
06	2050FCT	CH_015	10,000	15,000	**6,000**
06	2050FCT	CH_020	10,000	15,000	**15,000**
06	2050FCT	CH_010	10,000	**4,500**	30,000
06	2050FCT	CH_015	10,000	**3,000**	30,000
06	2050FCT	CH_020	10,000	**7,500**	30,000
06	2050FCT	CH_010	**3,000**	15,000	30,000
06	2050FCT	CH_015	**2,000**	15,000	30,000
06	2050FCT	CH_020	**5,000**	15,000	30,000

When allocating **GS Costs** , the system copies also the other measures instead put them to zero

Allocation of more measure - Solution

In the DTP, after each Allocation use an operation to put to zero the other measures

Period	Scenario	Channel	Marketing & Sales Allocated	IT Costs allocated	GS Costs allocated
06	2050FCT	CH_010	0	0	**9,000**
06	2050FCT	CH_015	0	0	**6,000**
06	2050FCT	CH_020	0	0	**15,000**
06	2050FCT	CH_010	0	**4,500**	0
06	2050FCT	CH_015	0	**3,000**	0
06	2050FCT	CH_020	0	**7,500**	0
06	2050FCT	CH_010	**3,000**	0	0
06	2050FCT	CH_015	**2,000**	0	0
06	2050FCT	CH_020	**5,000**	0	0

Add an operation to group by measures and have the expected result

Period	Scenario	Channel	Marketing & Sales Allocated	IT Costs allocated	GS Costs allocated
06	2050FCT	CH_010	**3,000**	**4,500**	**9,000**
06	2050FCT	CH_015	**2,000**	**3,000**	**6,000**
06	2050FCT	CH_020	**5,000**	**7,500**	**15,000**